Supplementary Information

GuardSynth: Checkable Behavioral Constraints for CoC-Augmented Candidate Selection

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This document supports the main manuscript. Sections are cited there as "Supplementary Section Sn". Rows and predictions are constructed synthetic examples rather than independent road scenes unless stated otherwise. Percent values are rounded half up to two decimals. Section S12 lists the archived records from which the reported values are read.

Metric definitions follow Section 4.2 of the main text. Preferred-selection score (PSS) is the percentage of predictions selecting the preferred admissible candidate; constraint violation rate (CVR) is the percentage selecting a prohibited candidate; rule-pair accuracy (RPA) requires both reference labels in a rule pair. Unextractable answers receive zero PSS credit and count as neither violations nor admissible choices. The label-match calculation and archived result fields are unchanged by this terminology.

S1 Motivating Alpamayo-R1 Diagnostics

These preliminary experiments use Alpamayo-R1-10B without additional training and assess sampled open-loop predictions under supplied experimental rules. They do not estimate executed driving violations, closed-loop safety, or a population-level road failure rate. Alpamayo-R1 (also named Alpamayo 1) is an earlier member of the model family than Alpamayo 1.5 evaluated in *Drive the Thoughts*; their weights and evaluations are distinct.

Inputs and rule routes. The speed test uses three NVIDIA PhysicalAI inputs and three inputs from YSHRobotics/CoC-Nusc, a third-party conversion of nuScenes to the PhysicalAI format. Each input is evaluated with three trajectory-generation seeds (42000, 43000, 44000). The user route adds the rule to the user prompt; the CoC route appends the supplied rule to the generated CoC continuation before trajectory generation. The latter is an intervention on the continuation, not a claim that the model generated the rule. Paired lower and upper limits share inputs and seeds.

Measurements. The lower limits are 8.6, 0.9, and 13.9 m/s for the PhysicalAI inputs and 5.2, 5.0, and 0.6 m/s for the converted nuScenes inputs; each is at least 0.5 m/s above the estimated current speed. The corresponding upper limits are 9.9, 9.9, and 21.9 m/s for PhysicalAI and 9.9 m/s for all three converted nuScenes inputs. Segment-average speed is computed from consecutive predicted XY positions at 0.1 s spacing, and a violation occurs when any segment in the first two seconds exceeds the limit by more than 0.05 m/s. The hold test uses one low-speed input per source and three seeds, requesting a position within 0.2 m of the current location for two seconds; a violation means a sampled location exceeds 0.21 m (0.01 m tolerance).

Maximum sampled hold displacement across seeds is 2.330/2.420 m for PhysicalAI and 1.856/1.655 m for

Table S1: Alpamayo-R1 diagnostic constraint violations (outputs; three seeds per input).

Source	Rule route	Speed violations	Hold violations
PhysicalAI	User prompt	6/9	3/3
PhysicalAI	CoC continuation	6/9	3/3
Converted nuScenes	User prompt	9/9	3/3
Converted nuScenes	CoC continuation	9/9	3/3

converted nuScenes (user/CoC routes). A 0.5 s hold test produced 3/3 violations per route for PhysicalAI and 0/3 for converted nuScenes. For each lower/upper-limit pair, the mean XY difference averages the distance between all 64 corresponding predicted positions over the full 6.4 s horizon, whereas violations are assessed over the first two seconds. The largest input-and-route mean over three seeds is 0.0868 m (converted nuScenes, user route); an individual trajectory-pair mean can reach 0.0962 m. Under the user route, the generated CoC did not include the supplied limit. Raw licensed images, videos, trajectories, and CoC outputs remain in restricted evidence packages; the converted source is YSHRobotics/CoC-Nusc (revision 994bdd94, CC BY-NC-SA 4.0), and NVIDIA PhysicalAI material retains its source restrictions.

S2 Data, Splits, and Training Details

Pedestrian entry. Training uses 40 synthetic images and 960 rows per condition (clearance times 6.0–9.9 s); validation and test each use 10 images and 240 rows (10.0–10.9 s and 11.0–11.9 s). Each image yields three candidate patterns, four label rotations, and two waiting rules. Images do not overlap across splits. Paraphrase and unseen-duration evaluations reuse test images; the latter uses waits of 0.8 and 1.8 s and shifts candidate entry times by 0.3 s.

Static and maneuver tasks. Static training has 576 rows, 96 scene IDs, and 18 images; the primary test has 144 rows, 24 scene IDs, and 13 images, all of which also occur in training. The eight unseen-value images do not occur in training. Maneuver training has 384 rows and 192 scene IDs, and each maneuver evaluation set has 96 rows and 48 scene IDs, using two repeated schematic images. The data seed is 20260930.

Position holding. Training, validation, and test contain 120, 16, and 32 profiles (960, 128, and 256 rows). Related rows stay in the same split.

Progress scores and labels. The pedestrian-entry generator assigns utilities 110, 100, 85, and 0 to ordered candidate templates; these are supplied synthetic scores, not values calculated from distance or speed. The position-hold generator varies score values across profiles and alternates the relative ranking of the candidate admissible for both durations and the candidate admissible only for the shorter duration; an additional candidate receives a lower score. In both tasks, A–D labels are cyclically reassigned to candidate behaviors.

Training. LoRA (rank 8, $\alpha = 16$, dropout 0.05) adapts the language-attention q/k/v/o projections of Qwen3-VL-2B-Instruct. Training uses bfloat16, AdamW (learning rate 2×10^{-4}, weight decay 0.01), batch size 2 with accumulation 8, three epochs, and loss on answer tokens only. The final checkpoint is used without validation- or test-based selection. Input lengths differ across text conditions, so token counts and FLOPs are not matched. Fresh-adapter attempts interrupted before model saving were restarted from the same base and seed; only completed fits enter the aggregates.

S3 Qualitative Lifecycle Specification

The qualitative specification interprets obligation j at decision step t using a condition status $truth[j,t] \in \{\text{TRUE}, \text{FALSE}, \text{UNKNOWN}, \text{CONFLICT}\}$, evidence acceptance $valid[j,t]$, and a Boolean obligation state $active[j,t]$. FALSE denotes confirmed resolution of the restricting condition, not a missing observation. Valid TRUE activates the obligation, valid FALSE releases it, and UNKNOWN, CONFLICT, or invalid evidence retains the previous state. The first decision uses a supplied initial state; later decisions use the preceding state.

Entry permission is separate from obligation state: $entry_permitted[t] = \bigwedge_j clear[j,t]$, where $clear[j,t]$ requires valid FALSE evidence for obligation j. Previous inactivity alone therefore does not permit entry. Starting inactive, the evidence sequence valid TRUE $\rightarrow$ UNKNOWN $\rightarrow$ FALSE $\rightarrow$ TRUE yields active $\rightarrow$ active $\rightarrow$ inactive $\rightarrow$ active, and entry is permitted only at the third decision. Unknown, conflicting, or

invalid evidence marks the case for review; these evidence values differ from a solver's UNKNOWN result.

The archived qualitative run contains 25 queries, all matching their expected verdicts. For example, if an obligation was active and the current evidence is UNKNOWN, asserting that it is now inactive must be UNSAT. These checks establish the specified update rules under supplied initial states and evidence sequences; they do not show that such sequences were correctly recovered from real-scene video. The learning experiments do not use this specification.

S4 Additional Specification Checks

Candidate-binding checks in the learning data. Across the training, validation, and test splits of the six-task pedestrian-entry data, 5,760 candidate-binding checks yield 3,840 SAT and 1,920 UNSAT outcomes; removing repeated assignments leaves 960 distinct assignments (540 SAT and 420 UNSAT). Of the 1,920 UNSAT checks, 480 bind entry before clearance and 1,440 bind entry after clearance but before the wait expires. Every outcome agrees with the separate integer rule, every row retains at least one admissible entry candidate, and every reference action is SAT. Static and maneuver candidate bindings comprise 5,088 checks (3,432 SAT and 1,656 UNSAT), and SMT agrees with the independent task rules on all 9,792 static and maneuver predictions. These counts are separate from the boundary queries and position-hold candidate checks in Section 5.9 of the main text.

Earlier fault injection on the expression-diversity specification. Before the evaluation of the learning specifications (Section 3.5 of the main text), faults were injected into a pedestrian-entry specification developed for the expression-diversity study. It supports waits from 0.1 to 3.0 s and records units and the inclusive comparison explicitly. Concrete probes combine entry or non-entry with times at -100, 0, and $+100$ ms around release. Table S2 reports the 192 cases; the fault types and detecting layers match those for the learning specifications.

Table S2: Fault injection into the expression-diversity specification.

Case	Detected / cases	Detecting layer
Changed sentence negation	24/24	Sentence correspondence
Changed sentence duration	24/24	Sentence correspondence
Deleted release condition	24/24	Logic–rule comparison
Reversed comparison	24/24	Logic–rule comparison
Changed specification duration	24/24	Evidence correspondence, logic–rule comparison
Changed unit	24/24	Structural check
Jointly wrong evidence and specification	0/24	Not detected
Valid specification	24/24 accepted	–

Bypassing the checks accepts all 192 cases; applying them accepts 48, namely the valid specifications and the joint-error controls. In the representative query with clearance at 11.000 s, a 0.500 s wait, and entry at 11.499 s, tracking assertions gives an UNSAT core with four members (the entry gate, clearance time, entry time, and entry flag); deleting any one makes the remainder SAT. Expected UNSAT for a prohibited candidate is retained as evidence of exclusion. Unexpected UNSAT, unexpected SAT, or UNKNOWN returns the specification for review of its source, threshold, and clauses, followed by rechecking consistency, exclusion, and post-release feasibility.

S5 Native Execution Checks of Related Systems

Table S3 records checks of the original execution paths of selected related methods. Released examples were run without adding our constraints or a candidate-selection adapter. These checks establish which native paths were exercised; they provide no shared-task ranking. Resource statements describe official sources inspected on 7 October 2026.

S6 Seed-Level and Detailed Results

Fresh-adapter study. Table S4 reports individual seeds with inputs matching each training condition, and Table S5 reports the CoC-trained adapters under common CNL. CNL-trained adapters receive the same

Table S3: Native execution checks.

Method	Native output	Check
nl2spec	Temporal logic with language–subformula links	One original CLI example executed with three translation samples; interactive correction not exercised
NL2TL	Temporal logic from supplied requirements	Eleven original notebook examples executed with the official weights
DriveLM	Graph VQA and driving reasoning	Not executed: the LLaMA-7B base weights required by the released challenge baseline were unavailable to us; this is not the full DriveLM-Agent
DriveVLM	Scene analysis and hierarchical planning	Not executed: the inspected official repository contained project-page material but no inference code or weights

input in both evaluations. The standard deviations in the main text are sample standard deviations over seeds ($n-1$ denominator). Each cell pools both durations: 240 pedestrian-entry rows or 128 position-hold rows per seed.

Table S4: Fresh adapters with training-matched inputs: preferred-selection score / constraint violation rate / rule-pair accuracy (%).

Task	Training	Seed 42	Seed 17	Seed 123
Pedestrian entry	CoC, CE	49.58 / 25.00 / 0.00	50.00 / 12.92 / 0.00	50.00 / 20.83 / 0.00
Pedestrian entry	CoC, CE + semantic	50.00 / 0.00 / 0.00	50.00 / 0.00 / 0.00	50.00 / 0.00 / 0.00
Pedestrian entry	CoC + CNL, CE	99.17 / 0.00 / 98.33	99.17 / 0.42 / 98.33	97.08 / 2.92 / 94.17
Pedestrian entry	CoC + CNL, CE + semantic	100.00 / 0.00 / 100.00	97.08 / 2.50 / 94.17	95.83 / 0.00 / 91.67
Hold, maxima	CoC, CE	75.00 / 11.72 / 50.00	75.00 / 9.38 / 50.00	75.00 / 11.72 / 50.00
Hold, maxima	CoC, CE + semantic	75.00 / 0.00 / 50.00	75.00 / 1.56 / 50.00	73.44 / 3.13 / 48.44
Hold, maxima	CoC + CNL, CE	97.66 / 0.00 / 95.31	95.31 / 0.00 / 90.63	95.31 / 0.00 / 92.19
Hold, maxima	CoC + CNL, CE + semantic	93.75 / 0.00 / 87.50	96.09 / 0.00 / 93.75	98.44 / 0.00 / 96.88
Hold, XY	CoC, CE	71.09 / 12.50 / 46.88	73.44 / 7.81 / 48.44	73.44 / 12.50 / 50.00
Hold, XY	CoC, CE + semantic	74.22 / 1.56 / 50.00	75.00 / 0.00 / 50.00	75.00 / 0.00 / 50.00
Hold, XY	CoC + CNL, CE	93.75 / 0.78 / 87.50	92.19 / 1.56 / 84.38	94.53 / 2.34 / 89.06
Hold, XY	CoC + CNL, CE + semantic	90.63 / 0.00 / 81.25	90.63 / 5.47 / 81.25	96.88 / 0.78 / 93.75

Table S5: CoC-trained fresh adapters with common CNL at evaluation: preferred-selection score / constraint violation rate / rule-pair accuracy (%).

Task	Training	Seed 42	Seed 17	Seed 123
Pedestrian entry	CoC, CE	48.75 / 25.00 / 0.83	51.67 / 7.92 / 5.00	50.00 / 20.83 / 0.00
Pedestrian entry	CoC, CE + semantic	50.00 / 0.00 / 0.00	50.00 / 0.00 / 0.00	50.00 / 0.00 / 0.00
Hold, maxima	CoC, CE	78.13 / 6.25 / 56.25	80.47 / 3.13 / 60.94	76.56 / 10.16 / 53.13
Hold, maxima	CoC, CE + semantic	76.56 / 0.00 / 53.13	75.00 / 0.78 / 51.56	75.00 / 0.00 / 50.00
Hold, XY	CoC, CE	75.78 / 4.69 / 51.56	75.78 / 4.69 / 51.56	75.00 / 9.38 / 51.56
Hold, XY	CoC, CE + semantic	74.22 / 1.56 / 50.00	75.00 / 0.00 / 50.00	75.00 / 0.00 / 50.00

Six-task study. For pedestrian entry, three-seed means ± sample standard deviations are: P0 49.86±0.24 preferred-selection score, 19.17 ± 7.11 constraint violation rate, 0.00 rule-pair accuracy; P1 96.81 ± 2.10, 1.39±2.06, 93.61±4.19; P2 98.06±1.88, 0.83±0.83, 96.11±3.76. These standard deviations use the unrounded per-seed rates in `temporal-expansion-learning-2026-09-29-001/RESULT.json`; the archived population-SD fields are unchanged. P2 does not dominate every seed; for example, seed 42 has no P1 violations and 0.83% P2 violations. Table S6 gives static and maneuver details, including constraint-compliant completion.

On the combined maneuver set, preferred-selection score / constraint violation rate / constraint-compliant completion / rule-pair accuracy are 50.00 / 30.21 / 69.79 / 0.00 for P0, 94.10 / 0.69 / 99.31 / 88.19 for P1 and 91.32 / 1.04 / 98.96 / 82.64 for P2. On the pedestrian-entry paraphrase, the alternative natural-language guard reaches 93.89 / 0.00 (preferred-selection score/constraint violation rate).

Individual seeds for the six-task and additional comparisons. Tables S7, S8, and S9 give the archived seed values underlying main-text means. Seeds follow the protocol order 42, 17, and 123. Primary subtask scores and family-level robustness scores have different aggregation levels; their rows are not additional independent evaluations. Pedestrian-entry rows use 240 predictions per seed, static subtasks 48, maneuver subtasks 48, static family evaluations 144, and maneuver family evaluations 96.

Continuation study. Table S10 gives preferred-candidate selections before and after continuation (192

Table S6: Static and maneuver results: preferred-selection score / constraint violation rate / constraint-compliant completion / rule-pair accuracy (%), three-seed means.

Family	Condition	Primary	Paraphrase	Unseen values
Static	P0	50.00/24.31/75.69/0.00	50.00/24.31/75.69/0.00	50.00/22.92/77.08/0.00
Static	P1	100.00/0.00/100.00/100.00	93.29/0.00/100.00/86.57	99.54/0.00/100.00/99.07
Static	P2	100.00/0.00/100.00/100.00	80.32/7.87/92.13/60.65	97.22/0.23/99.77/94.44
Maneuver	P0	50.00/30.90/69.10/0.00	50.00/30.90/69.10/0.00	50.00/30.90/69.10/0.00
Maneuver	P1	100.00/0.00/100.00/100.00	100.00/0.00/100.00/100.00	100.00/0.00/100.00/100.00
Maneuver	P2	98.96/0.00/100.00/97.92	97.92/2.08/97.92/95.83	98.96/0.00/100.00/97.92

Table S7: Six-task primary results by individual seed: preferred-selection score / constraint violation rate / rule-pair accuracy (%).

Task	Condition	Seed 42	Seed 17	Seed 123
Pedestrian entry	P0	49.58 / 25.00 / 0.00	50.00 / 11.25 / 0.00	50.00 / 21.25 / 0.00
Pedestrian entry	P1	97.08 / 0.00 / 94.17	94.58 / 3.75 / 89.17	98.75 / 0.42 / 97.50
Pedestrian entry	P2	96.25 / 0.83 / 92.50	100.00 / 0.00 / 100.00	97.92 / 1.67 / 95.83
Maximum speed	P0	50.00 / 22.92 / 0.00	50.00 / 25.00 / 0.00	50.00 / 25.00 / 0.00
Maximum speed	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Maximum speed	P2	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Minimum clearance	P0	50.00 / 22.92 / 0.00	50.00 / 25.00 / 0.00	50.00 / 25.00 / 0.00
Minimum clearance	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Minimum clearance	P2	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Minimum entry delay	P0	50.00 / 22.92 / 0.00	50.00 / 25.00 / 0.00	50.00 / 25.00 / 0.00
Minimum entry delay	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Minimum entry delay	P2	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Stopping	P0	50.00 / 20.83 / 0.00	50.00 / 25.00 / 0.00	50.00 / 25.00 / 0.00
Stopping	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Stopping	P2	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Lane change	P0	50.00 / 25.00 / 0.00	50.00 / 39.58 / 0.00	50.00 / 50.00 / 0.00
Lane change	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Lane change	P2	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	93.75 / 0.00 / 87.50

per cell).

Table S8: Pedestrian-entry representations and robustness by individual seed: preferred-selection score / constraint violation rate / rule-pair accuracy (%).

Evaluation	Condition	Seed 42	Seed 17	Seed 123
Primary	P0	49.58 / 25.00 / 0.00	50.00 / 11.25 / 0.00	50.00 / 21.25 / 0.00
Primary	P1	97.08 / 0.00 / 94.17	94.58 / 3.75 / 89.17	98.75 / 0.42 / 97.50
Primary	P2	96.25 / 0.83 / 92.50	100.00 / 0.00 / 100.00	97.92 / 1.67 / 95.83
Primary	Alternative guard	95.83 / 3.33 / 91.67	99.17 / 0.00 / 98.33	99.17 / 0.00 / 98.33
Primary	Logical text	98.33 / 0.00 / 96.67	99.17 / 0.42 / 98.33	100.00 / 0.00 / 100.00
Paraphrase	P0	49.58 / 25.00 / 0.00	50.00 / 11.25 / 0.00	50.00 / 21.25 / 0.00
Paraphrase	P1	87.92 / 7.92 / 75.83	87.50 / 6.67 / 75.00	96.25 / 0.42 / 92.50
Paraphrase	P2	93.33 / 1.25 / 86.67	99.17 / 0.83 / 98.33	98.33 / 1.67 / 96.67
Paraphrase	Alternative guard	92.08 / 0.00 / 84.17	90.00 / 0.00 / 80.00	99.58 / 0.00 / 99.17
Paraphrase	Logical text	98.33 / 0.00 / 96.67	99.17 / 0.42 / 98.33	100.00 / 0.00 / 100.00
Unseen wait	P0	49.58 / 25.00 / 0.00	50.00 / 9.58 / 0.00	50.00 / 20.83 / 0.00
Unseen wait	P1	98.75 / 0.00 / 97.50	92.92 / 6.25 / 85.83	98.33 / 0.42 / 96.67
Unseen wait	P2	94.58 / 0.42 / 89.17	99.17 / 0.83 / 98.33	96.25 / 3.33 / 92.50
Unseen wait	Alternative guard	97.08 / 1.67 / 94.17	99.17 / 0.00 / 98.33	99.58 / 0.00 / 99.17
Unseen wait	Logical text	96.67 / 0.00 / 93.33	97.08 / 2.50 / 94.17	50.00 / 0.00 / 0.00

Table S9: Static and maneuver comparisons by individual seed: preferred-selection score / constraint violation rate / rule-pair accuracy (%). Family values aggregate their subtasks.

Family, evaluation	Condition	Seed 42	Seed 17	Seed 123
Static, primary	P0	50.00 / 22.92 / 0.00	50.00 / 25.00 / 0.00	50.00 / 25.00 / 0.00
Static, primary	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Static, primary	P2	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Static, paraphrase	P0	50.00 / 22.92 / 0.00	50.00 / 25.00 / 0.00	50.00 / 25.00 / 0.00
Static, paraphrase	P1	93.06 / 0.00 / 86.11	91.67 / 0.00 / 83.33	95.14 / 0.00 / 90.28
Static, paraphrase	P2	84.72 / 6.94 / 69.44	80.56 / 8.33 / 61.11	75.69 / 8.33 / 51.39
Static, unseen	P0	50.00 / 18.75 / 0.00	50.00 / 25.00 / 0.00	50.00 / 25.00 / 0.00
Static, unseen	P1	100.00 / 0.00 / 100.00	98.61 / 0.00 / 97.22	100.00 / 0.00 / 100.00
Static, unseen	P2	97.92 / 0.00 / 95.83	99.31 / 0.69 / 98.61	94.44 / 0.00 / 88.89
Maneuver, primary	P0	50.00 / 22.92 / 0.00	50.00 / 32.29 / 0.00	50.00 / 37.50 / 0.00
Maneuver, primary	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Maneuver, primary	P2	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	96.88 / 0.00 / 93.75
Maneuver, paraphrase	P0	50.00 / 22.92 / 0.00	50.00 / 32.29 / 0.00	50.00 / 37.50 / 0.00
Maneuver, paraphrase	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Maneuver, paraphrase	P2	100.00 / 0.00 / 100.00	93.75 / 6.25 / 87.50	100.00 / 0.00 / 100.00
Maneuver, unseen	P0	50.00 / 25.00 / 0.00	50.00 / 28.13 / 0.00	50.00 / 39.58 / 0.00
Maneuver, unseen	P1	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00
Maneuver, unseen	P2	100.00 / 0.00 / 100.00	100.00 / 0.00 / 100.00	96.88 / 0.00 / 93.75
Maneuver, combined	P0	50.00 / 25.00 / 0.00	50.00 / 28.13 / 0.00	50.00 / 37.50 / 0.00
Maneuver, combined	P1	87.50 / 0.00 / 75.00	96.88 / 0.00 / 93.75	97.92 / 2.08 / 95.83
Maneuver, combined	P2	89.58 / 0.00 / 79.17	98.96 / 1.04 / 97.92	85.42 / 2.08 / 70.83

Table S10: Position-hold continuation: preferred-candidate selections before → after training (of 192).

Representation, hold	P0	P1	P2
Maxima, 0.5 s	92 → 129	108 → 172	100 → 182
Maxima, 2.0 s	121 → 167	165 → 189	157 → 189
XY, 0.5 s	92 → 119	99 → 156	96 → 174
XY, 2.0 s	63 → 169	65 → 175	76 → 188

S7 Pretrained-VLM Evaluation

Both pretrained models are evaluated once per input with greedy decoding. The 12-token limit was fixed before execution; the 512-token evaluation with final-answer extraction was added after truncated explanations were observed. Both are reported. Unextractable answers count as incorrect and as neither violations nor admissible choices. Table S11 gives all results. Rule-pair accuracy of the 8B model at the 12-token limit, recorded in the evaluation results with the same pair definition as for the fine-tuned adapters, is 0.83% for pedestrian entry, 9.72% for the static tasks and 25.00% for the maneuver tasks (means of their subtasks), and 51.56% and 23.44% for interval-maximum and XY position holding; an unextractable answer counts as incorrect.

Table S11: Pretrained models with CoC + CNL inputs: preferred-selection score / constraint violation rate (%), answer-extraction failures (%), and output-limit hits.

Model	Limit	Task	PSS / CVR	Extraction failures	Limit hits
2B	12	Pedestrian entry	0.00 / 0.00	100.00	240/240
2B	12	Maximum speed	0.00 / 0.00	100.00	48/48
2B	12	Minimum clearance	0.00 / 0.00	100.00	48/48
2B	12	Minimum entry delay	0.00 / 0.00	100.00	48/48
2B	12	Stopping	0.00 / 0.00	100.00	48/48
2B	12	Lane change	0.00 / 0.00	100.00	48/48
2B	12	Hold, maxima	0.00 / 0.00	100.00	128/128
2B	12	Hold, XY	0.00 / 0.00	100.00	128/128
2B	512	Pedestrian entry	10.83 / 11.25	65.42	240/240
2B	512	Maximum speed	50.00 / 50.00	0.00	0/48
2B	512	Minimum clearance	50.00 / 50.00	0.00	0/48
2B	512	Minimum entry delay	54.17 / 14.58	31.25	15/48
2B	512	Stopping	0.00 / 0.00	100.00	48/48
2B	512	Lane change	37.50 / 8.33	54.17	26/48
2B	512	Hold, maxima	14.84 / 0.00	79.69	112/128
2B	512	Hold, XY	7.03 / 7.03	75.78	128/128
8B	12	Pedestrian entry	27.08 / 68.75	0.00	0/240
8B	12	Maximum speed	60.42 / 39.58	0.00	0/48
8B	12	Minimum clearance	54.17 / 45.83	0.00	0/48
8B	12	Minimum entry delay	50.00 / 50.00	0.00	0/48
8B	12	Stopping	50.00 / 25.00	0.00	0/48
8B	12	Lane change	75.00 / 0.00	0.00	0/48
8B	12	Hold, maxima	70.31 / 7.03	0.00	0/128
8B	12	Hold, XY	35.94 / 33.59	20.31	26/128
8B	512	Pedestrian entry	27.50 / 66.67	0.00	0/240
8B	512	Maximum speed	60.42 / 39.58	0.00	0/48
8B	512	Minimum clearance	54.17 / 45.83	0.00	0/48
8B	512	Minimum entry delay	50.00 / 50.00	0.00	0/48
8B	512	Stopping	50.00 / 25.00	0.00	0/48
8B	512	Lane change	75.00 / 0.00	0.00	0/48
8B	512	Hold, maxima	70.31 / 7.03	0.00	0/128
8B	512	Hold, XY	36.72 / 34.38	17.19	29/128

S8 General Reasoning-Prompt Controls

Zero-shot CoT appends its step-by-step trigger and then extracts an answer. Basic Plan-and-Solve uses the original planning-and-solving prompt, extracting directly when the specified answer marker is present and otherwise generating a second extraction response. Neither PS+ nor self-consistency is used. Reasoning limits are 128 tokens for Zero-shot CoT and 256 for Plan-and-Solve, with up to 32 tokens for extraction. With greedy decoding, identical image-and-prompt inputs within a method and checkpoint reuse one output: 120, 24, and 38 unique inputs per family, giving 1,092 first-stage generations. Each method scores 1,440 predictions (2,880 in total). All 546 unique Zero-shot CoT first-stage outputs and 498 of 546 Plan-and-Solve outputs are short label strings; the other 48 Plan-and-Solve outputs, all static, reach the 256-token limit before extraction. Table S12 gives task-level results.

Table S12: Reasoning-prompt controls on CoC-only checkpoints: preferred-selection score / constraint violation rate (%), three-seed means. Rule-pair accuracy is 0% throughout.

Task	P0	Zero-shot CoT	Plan-and-Solve
Pedestrian entry	49.86 / 19.17	49.86 / 18.19	49.86 / 17.92
Maximum speed	50.00 / 24.31	50.00 / 25.00	50.00 / 25.69
Minimum clearance	50.00 / 24.31	50.00 / 25.00	50.00 / 25.69
Minimum entry delay	50.00 / 24.31	50.00 / 25.00	50.00 / 25.69
Stopping	50.00 / 23.61	50.00 / 23.61	50.00 / 25.00
Lane change	50.00 / 38.19	50.00 / 36.81	50.00 / 35.42

S9 Recorded Prediction Cases

Two cases were selected from stored seed-42 test predictions by sorting records by scene ID and rule and taking (a) the first case in which P0 violates while P1 and P2 are correct and (b) the first P2 violation. They are explanatory, not representative samples. In case (a), clearance occurs at 11.0 s and a 1.5 s wait permits entry from 12.5 s; P0 selects entry at 11.5 s, whereas P1 and P2 select the reference entry at 12.5 s. In case (b), clearance occurs at 11.4 s and entry is permitted from 12.9 s; P0 and P1 select the reference candidate, whereas P2 enters at 12.8 s, 0.1 s early. Labels rotate, so a letter does not identify a fixed behavior.

S10 Follow-Up Diagnostics of Pedestrian-Entry Adapters

Nine existing P0, P1, and P2 adapters produce 2,160 exploratory predictions without new training (Table S13). D1 changes boundaries, clearance instants, waits, or their combination; D2 changes wording; D3 uses original, blank, or redrawn images. Each bundle contains three semantic parameter groups and 24 rows per condition and seed. For D3, a group is a base case configuration evaluated with two waiting rules and four label rotations. Redrawn images alter the visual premise, so these preferred-selection scores are input-intervention results. P2 changes its choice on 25.00% of blank-image and 19.44% of redrawn-image cases, and P1 on 18.06% and 12.50%.

Table S13: Follow-up diagnostics: preferred-selection score / constraint violation rate (%), three-seed means.

Diagnostic	P0	P1	P2
D1, boundary	50.00 / 33.33	95.83 / 4.17	95.83 / 2.78
D1, clearance only	50.00 / 22.22	100.00 / 0.00	98.61 / 1.39
D1, duration only	50.00 / 20.83	98.61 / 1.39	97.22 / 2.78
D1, joint	50.00 / 22.22	100.00 / 0.00	98.61 / 1.39
D2, canonical	50.00 / 20.83	100.00 / 0.00	95.83 / 2.78
D2, paraphrase 1	50.00 / 20.83	98.61 / 0.00	97.22 / 2.78
D2, paraphrase 2	50.00 / 20.83	76.39 / 15.28	77.78 / 9.72
D3, original image	50.00 / 22.22	100.00 / 0.00	98.61 / 1.39
D3, blank image	50.00 / 13.89	81.94 / 2.78	76.39 / 1.39
D3, redrawn image	50.00 / 19.44	87.50 / 0.00	79.17 / 1.39

S11 Position-Hold Regression Diagnostic

The regression set reuses 256 previously inspected diagnostic rows, separate from the new test set. Table S14 gives strict violations, tolerance-based violations (0.01 m) in parentheses, and preferred-selection score after continuation. Before continuation, P2 had 75/192 strict violations in the XY, 2.0 s stratum. The smallest prohibited overshoot is 1 mm here, compared with 22 mm in the new test.

S12 Data Records

Results are read from frozen evidence packages under `projects/04-guardsynth-coc/artifacts/public/`:

- `paper1-vlm-constraint-baselines-001`: fresh-adapter study and pretrained models;
- `paper1-position-hold-training-001`: continuation study and regression diagnostic;

Table S14: Regression diagnostic after continuation: strict (tolerance) violations of 192 and preferred-selection score (%).

	P0		P1		P2	
Representation, hold	Viol.	PSS	Viol.	PSS	Viol.	PSS
Maxima, 0.5 s	0 (0)	80.21	0 (0)	90.63	0 (0)	96.88
Maxima, 2.0 s	53 (18)	70.83	16 (1)	89.58	1 (0)	98.44
XY, 0.5 s	5 (5)	81.77	17 (17)	79.69	40 (39)	75.00
XY, 2.0 s	106 (53)	41.67	79 (41)	58.85	77 (50)	59.38

- `paper1-learning-specification-faults-001`: fault injection into learning specifications;
- `paper1-constraint-composition-001/` contains two runs:

  `document-grounded-composition-2026-10-17-002`: document-grounded diagnostic;

  `constraint-composition-2026-10-17-001`: synthetic boundary diagnostic. Both are reported separately in Section S13.
- `paper1-method-preservation-001` and `paper1-temporal-diversity-001`: six-task study, representations, diagnostics, and specification checks;
- `paper1-reasoning-comparison-001`: reasoning-prompt controls;
- `paper1-external-methods-001/original-method-checks-2026-10-07-001/RESULT.json`: native execution checks in Section S5.

Restricted records are under `projects/04-guardsynth-coc/artifacts/restricted/`:

- `alpamayo-speed-limits-001/speed-limits-2026-10-07-001/RESULT.json`: PhysicalAI speed diagnostics;
- `alpamayo-nuscenes-speed-limits-001/speed-limits-2026-10-08-001/RESULT.json`: converted nuScenes speed diagnostics;
- `alpamayo-position-hold-001/position-hold-2026-10-08-001/RESULT.json`: position-hold diagnostics for both sources;
- `guardsynth-paper1-action-contract-001/scene18-action-contract-2026-09-08-002/query_checks.json`: qualitative lifecycle queries in Section S3.

The speed runs also contain `compliance_metrics.csv` and `rule_pair_metrics.csv`; the position-hold run contains `hold_metrics.csv`. Licensed source material remains in these restricted packages.

S13 Bounded Source-Composition Diagnostic

Document-grounded compatibility diagnostic

The current main-text diagnostic uses manually interpreted clauses, not source documents generated for the experiment. Its source catalog records versions, official URLs, page/paragraph locators, reviewer paraphrases, selected roles, and exclusions. Eight official documents were inspected: NTSB's Tempe report, two FHWA signal publications, Waymo's June 2026 Jaguar guide, Highway Code Rules 129 and 219, the Cruise recall report, and 49 CFR 392.10. The last three inform scope rather than executable cases: Cruise's full response guard is undisclosed; Rule 219 requires compliance with traffic signs; the railway stopping requirement has vehicle-class limits and exceptions. Two UNECE candidates (Regulations 152 and 157) were excluded from numerical models because direct official-PDF retrieval was blocked. Drafts or secondary copies were not substituted.

The catalog stores manually reviewed metadata and paraphrases, not byte-preserved original PDFs. Scenario states are authored. The natural statements in Table S15 explain the formalization; their automatic extraction and parse-back are not evaluated.

The platform compiler and solver are unchanged. Variables use integer ms and mm as dimensionless counts; inequalities are cross-multiplied to preserve exact rational stopping distances. Stop positions are conservatively rounded upward to the mm grid.

Table S15: Document-grounded conditions and authored inputs for composition checking. Formulas use SI units; execution uses integer ms/mm and exact cross-multiplication. Scenario values are hypothetical, not measured accident states. b denotes choosing the stop branch; m autonomous driving; c crossing; q, n, f a stationary vehicle, necessity, and a clear road; o, p, k, a_d an open door, Park, applied parking brake, and deployed airbag. The faithful and omitted-exception encodings are alternatives, not simultaneous requirements.

Source / role	Natural-language statement	Formal condition
Braking: $t_b, x \geq 0$; selected emergency branch.		
NTSB policy record	The selected historical policy suppresses braking for 1 s.	$t_b \geq 1$
Authored motion model	Constant speed before braking; constant deceleration a thereafter.	$x \geq vt_b + v^2/(2a)$
Authored test objective	Stop within the supplied distance d.	$x \leq d$
Yellow interval: only stop or constant-speed proceed.		
FHWA-informed stopping model	The stop branch needs reaction and stopping distance.	$b \Rightarrow d \geq v\rho + v^2/(2a)$
FHWA-informed phase model	The proceed branch reaches the stop line by yellow end.	$\neg b \Rightarrow d \leq vY$
Autonomous-driving interlock: authored vehicle-state flags.		
Waymo operating guide	Any listed blocker prevents autonomous driving.	$(o \vee p \vee k \vee a_d) \Rightarrow \neg m$
Authored candidate	Test a candidate that drives autonomously.	$m = \top$
Rule 129: nearer line solid; all other exceptions absent.		
Rule 129, restricted	Crossing requires the stationary-vehicle exception's conditions.	$c \Rightarrow (q \wedge n \wedge f)$
Omitted-exception fault	Faulty alternative: prohibit crossing unconditionally.	$\neg c$
Authored candidate	Test crossing; separate controls hold instead.	$c = \top$

For braking, the selected historical emergency branch requires a persistent hazard, no takeover, and no alternative path. Emergency-trigger eligibility is not reconstructed; the original trigger involved required deceleration above 7 m/s^2 or jerk magnitude above 5 m/s^3. The constant-deceleration model ignores jerk, so it is not the original vehicle dynamics. NTSB reports that avoidance already exceeded ADS braking specifications at recognition in the actual collision; we make no counterfactual claim about preventing Tempe.

For $v \in \{10, 15, 20\}$ m/s, $a \in \{5, 7\}$ m/s^2, and $\delta \in \{0, 1\}$ s, distances bracket $v\delta + v^2/(2a)$ by one integer mm, after upward rounding. Each speed/deceleration pair also has a distance halfway between immediate and one-second-delayed stopping, tested under both delays (48 cases). Zero delay is an authored control. Distances shorter than immediate stopping remain in the grid to distinguish already-infeasible avoidance from delay incompatibility.

Signal cases fix $\rho = 1$ s and nominal $a = 3$ m/s^2, a rounded value below FHWA's historical 10 ft/s^2 guidance, with speeds 10/15/20 m/s and yellow 3/4/5 s. For each stopping and stop-line-reaching threshold z in mm, distances include $\lfloor z \rfloor - 1$, $\lceil z \rceil$, and $\lceil z \rceil + 1$, plus the floored midpoint of the thresholds; the 15 m/s, 3 s case also tests 50 m. Duplicate distances within a parameter setting are removed (64 cases). Only nominal stopping and unchanged-speed proceeding are modeled. The yellow endpoint uses a closed mathematical boundary, not a claim about legal permission at the exact instant red begins. We test reaching the stop line, not complete intersection clearance. For example, $v = 15$ m/s, $\rho = 1$ s, and $a = 3$ m/s^2 require 52.5 m for nominal stopping, whereas 3 s of unchanged-speed travel reaches 45 m. At 50 m from the stop line, neither branch is feasible; a 4 s interval admits proceeding. This is feasibility under the nominal two-action model, not impossibility of emergency braking or another manoeuvre.

The interlock grid tests all 16 boolean combinations of door-open, Park, parking-brake-applied, and airbag-deployed states, each with an autonomous-drive or hold candidate (32 cases). Rolling and towing are outside this interlock. The exception grid tests the eight states of stationary vehicle, necessity, and clear road, each with cross/hold and faithful/omitted-exception variants (32 cases). Other Rule 129 exceptions are fixed absent. The omitted-exception variant is an intentional transcription fault, not the published rule.

Every subset is solved with an independent feasibility reference. All 896 verdicts agree, all 832 SAT witnesses pass numeric checks, and UNKNOWN outcomes are zero. The 58 UNSAT full configurations have subset-minimal groups. Twelve braking cases require all three groups; six others already have an incompatible model/objective pair without the delay constraint. The 16 interlock hold controls and the 16 non-crossing controls are SAT. The faithful rule accepts crossing when all exception conditions hold, whereas the omitted-exception variant rejects it. These counts are separate from learning, fault injection, and the synthetic diagnostic below.

The maintained runner and protocol are `document_grounded_composition.py` and

Table S16: Independent full-combination feasibility references for the document-grounded grid, in SI units. δ is delay, B any active interlock blocker, E the three exception conditions together, and F the faithful encoding.

Family	Full composition is SAT iff	Cases	Queries
Braking	$v\delta + v^2/(2a) \le d$	48	384
Yellow	$d \ge v\rho + v^2/(2a) \vee d \le vY$	64	256
Interlock	$\neg m \vee \neg B$	32	128
Exception	$\neg c \vee (F \wedge E)$	32	128
Total		176	896

`document_grounded_composition_protocol_v0_1.json` in `projects/04-guardsynth-coc/experiments/paper1_cnl_learning/`. The frozen run is `document-grounded-composition-2026-10-17-002` under `projects/04-guardsynth-coc/artifacts/public/paper1-constraint-composition-001/`. It contains `SOURCE_CATALOG.json`, cases, subset queries with SMT-LIB and witnesses, minimal conflict groups, independent-reference results, and input/output hashes. `REPORT_KO.md` includes natural/formal tables and the source-selection rationale.

Earlier synthetic boundary diagnostic

This earlier logical experiment uses supplied synthetic source records S1–S9 listed in its frozen case records and report. Each source has its own ID and a predefined policy, operator-requirement, or modeling-assumption role. These are authored records in the protocol, not retrieved regulations or documents from actual organizations. The natural statements are explanatory templates instantiated from structured parameters; their translation or composition as CNL is not evaluated.

The maintained protocol and runner are in `projects/04-guardsynth-coc/experiments/paper1_cnl_learning/`. Their filenames are `constraint_composition_protocol_v0_1.json` and `constraint_composition.py`. The entry and hold components call the unchanged learning-profile lowerers. The platform Core compiler and Z3 check the shared context alone and every nonempty subset of the two or three source groups. No VLM inference or training is involved.

The analytic feasibility reference for the full combination is given below. Every proper subset is feasible under the fixed parameter grid; each such verdict is also solved and checked. A SAT witness is independently checked against the numerical conditions, without interpreting the Core expressions.

Table S17: Full-combination feasibility references and query counts. d is the entry deadline; τ is clearance time; Δ is waiting duration; t_d is the departure sample time; H is hold duration; D is required displacement or distance; V is the speed cap; T is travel duration; and δ is the minimum release-to-entry delay.

Composition	Full combination is SAT iff	Cases	Queries
Entry wait + deadline	$\neg e \vee d \ge \tau + \Delta$	48	192
Position hold + departure	$t_d > H \vee D \le 200$	12	48
Speed cap + progress	$D \le VT$	27	108
Three-way timing	$d \ge \Delta + \delta$	36	288
Total		123	636

The frozen protocol specifies the complete original grid; the main text now emphasizes the separate document-grounded diagnostic described above. The entry cases include 12 controls without an entry obligation; these retain SAT despite incompatible bounds conditional on entry. Hold cases include the inclusive endpoint and the next listed post-release sample, separating a genuine simultaneous conflict from a permitted later departure. The speed cases assume nonnegative constant speed, and hold cases assume nonnegative longitudinal coordinates and zero lateral coordinates at listed samples. No interpolation, acceleration, or vehicle dynamics are inferred.

All 123 contexts and individual groups are satisfiable. The 35 incompatible full combinations have subset-minimal conflict groups relative to the fixed context. Component-only checks identify none; pairwise checks identify 23, leaving 12 three-way conflicts with satisfiable pairs. Moving each conflicting threshold by one

unit to its equality boundary gives 35 solved SAT controls. All 636 verdicts match the reference; all 601 SAT witnesses pass independent numeric checks, and UNKNOWN outcomes are zero. These counts are not pooled with the fault-injection cases, queries, or model predictions.

The frozen run in Section S12 contains `cases.jsonl` (source records, natural statements, formal conditions, and contexts), `queries.jsonl` (selected sources, verdicts, SMT-LIB, and witnesses), and `compositions.jsonl` (Core models and minimal conflict groups). `RESULT.json` reports aggregate and per-family counts; `RUN_MANIFEST.json` records source hashes and the solver runtime. `REPORT_KO.md` includes the source-by-source natural/formal tables. An UNSAT group diagnoses incompatibility under the supplied context, not the correctness of source evidence, which requirement should take priority, or whether an actual driving system is safe.